

COCONUT OIL - CONFECTIONERY GRADE SPECIFICATION

PRODUCT INFORMATION

Product Name: Refined Bleached Deodorized (RBD) Coconut Oil - Confectionery Grade

Product Code: CO-CF-001

Grade: Food Grade - Confectionery Application

Form: Solid at room temperature, liquid above 24°C

Package: 20kg cartons, 200kg drums, bulk containers

PHYSICAL & CHEMICAL PROPERTIES

Basic Specifications

- **Free Fatty Acids (as Lauric):** Maximum 0.1%
- **Peroxide Value:** Maximum 3.0 meq O₂/kg
- **Moisture & Impurities:** Maximum 0.1%
- **Iodine Value:** 6-11 (Wijs method)
- **Saponification Value:** 250-260 mg KOH/g
- **Unsaponifiable Matter:** Maximum 0.5%

Color & Appearance

- **Lovibond Color (5¼" cell):** Maximum 1.0 Red, 10 Yellow
- **Appearance:** Crystal clear when liquid, white when solid
- **Odor:** Bland, practically odorless
- **Taste:** Bland, neutral taste profile

Melting Characteristics

- **Melting Point:** 24-26°C (75-79°F)
- **Slip Melting Point:** 23-25°C
- **Cloud Point:** 20-22°C
- **Solid Fat Content (SFC):**
 - At 10°C: 80-90%
 - At 15°C: 70-80%
 - At 20°C: 50-65%
 - At 25°C: 5-15%
 - At 30°C: 0-2%

FATTY ACID COMPOSITION

Typical Fatty Acid Profile (%)

- **Caproic Acid (C6:0):** 0.4-0.6%
- **Caprylic Acid (C8:0):** 5.5-9.5%
- **Capric Acid (C10:0):** 4.5-8.0%
- **Lauric Acid (C12:0):** 45-52%
- **Myristic Acid (C14:0):** 16-21%
- **Palmitic Acid (C16:0):** 7.5-10.5%
- **Stearic Acid (C18:0):** 2.0-4.0%
- **Oleic Acid (C18:1):** 5.0-8.0%
- **Linoleic Acid (C18:2):** 1.0-2.5%

Saturated vs Unsaturated

- **Total Saturated Fatty Acids:** 92-95%
- **Total Unsaturated Fatty Acids:** 5-8%
- **Medium Chain Fatty Acids (C6-C12):** 55-65%
- **Trans Fatty Acids:** <0.1% (naturally occurring)

CONFECTIONERY APPLICATIONS

Primary Uses

- **Chocolate Alternatives:** Base fat for compound coatings and white chocolate
- **Premium Chocolate:** Cocoa butter alternative in high-end products
- **Coating Applications:** Confectionery coatings and enrobing
- **Filled Chocolates:** Centers and ganaches
- **Cookies & Biscuits:** Texture enhancement and shelf-life extension
- **Cake Icings:** Stability and gloss in buttercream and fondant
- **Caramels & Fudge:** Texture modifier and processing aid
- **Coconut-Based Confections:** Authentic coconut flavor products

Functional Properties

- **Sharp Melting Profile:** Quick, clean melt in mouth at body temperature
- **Excellent Stability:** High resistance to oxidation and rancidity
- **Good Snap:** Provides crisp texture at room temperature

- **Flavor Enhancement:** Subtle coconut notes (in less refined grades)
- **Bloom Resistance:** Excellent resistance to fat bloom
- **Temperature Tolerance:** Stable in tropical climates
- **Natural Preservation:** Antimicrobial properties extend shelf life

Processing Characteristics

- **Easy Processing:** Simple melting and handling
- **Quick Setting:** Rapid crystallization for fast production cycles
- **Tempering Properties:** Excellent for chocolate tempering
- **Compatibility:** Blends well with other tropical fats
- **No Hydrogenation:** Naturally solid without trans fats
- **Heat Stability:** Stable at confectionery processing temperatures

NUTRITIONAL INFORMATION (per 100g)

Energy & Macronutrients

- **Energy:** 862 kcal (3605 kJ)
- **Total Fat:** 100g
- **Saturated Fat:** 92g
- **Monounsaturated Fat:** 6g
- **Polyunsaturated Fat:** 2g
- **Trans Fat:** <0.1g
- **Cholesterol:** 0mg

Micronutrients

- **Vitamin E (Tocopherols):** 0.1-0.5mg
- **Vitamin K:** Trace amounts
- **Phytosterols:** 80-120mg

Medium Chain Triglycerides (MCTs)

- **MCT Content:** 55-65% of total fatty acids
- **Lauric Acid Benefits:** Antimicrobial properties
- **Quick Energy Release:** Rapid metabolism characteristics

QUALITY PARAMETERS

Contaminants & Residues

- **Heavy Metals:**
 - Lead (Pb): <0.1 ppm
 - Cadmium (Cd): <0.05 ppm
 - Mercury (Hg): <0.05 ppm
 - Arsenic (As): <0.1 ppm
- **Pesticide Residues:** Below MRL limits
- **Aflatoxins:** <4 µg/kg (B1+B2+G1+G2)
- **PAH (Benzo[a]pyrene):** <2.0 µg/kg

Microbiological Specifications

- **Total Plate Count:** <100 cfu/g
- **Yeasts & Molds:** <100 cfu/g
- **E. coli:** Negative/g
- **Salmonella:** Negative/25g
- **Coliforms:** <10 cfu/g
- **Pathogenic Bacteria:** Absent

STORAGE & HANDLING

Storage Conditions

- **Temperature:** 15-25°C (59-77°F) for solid storage
- **Temperature:** Up to 50°C (122°F) for liquid storage
- **Relative Humidity:** <70%
- **Storage Life:** 24 months from production date
- **Light Protection:** Store in cool, dark place
- **Container:** Airtight containers to prevent moisture absorption

Handling Requirements

- **Material Compatibility:** Stainless steel, food-grade plastics
- **Avoid:** Copper, brass, iron (promote oxidation)
- **Gentle Heating:** Low temperature heating for liquefaction
- **Filtration:** Fine filtration for premium applications
- **Contamination Prevention:** Maintain strict hygiene protocols

VARIETIES & GRADES

Refining Levels

- **RBD (Refined, Bleached, Deodorized):** Neutral taste and odor
- **Virgin Coconut Oil:** Mild coconut flavor and aroma
- **Organic Coconut Oil:** Certified organic production
- **Fractionated Coconut Oil:** Modified melting characteristics

Specialty Grades

- **Confectionery Grade:** Optimized for chocolate and confectionery
- **Pharmaceutical Grade:** High purity for nutraceutical applications
- **Kosher/Halal Certified:** Religious dietary compliance
- **Fair Trade Certified:** Ethical sourcing certification

SUSTAINABILITY & CERTIFICATION

Sustainability Standards

- **Organic Certification:** USDA Organic, EU Organic
- **Fair Trade Certification:** Fair trade sourcing programs
- **Rainforest Alliance:** Sustainable farming practices
- **Carbon Footprint:** Lower than many alternative fats

Quality Certifications

- **HACCP:** Hazard Analysis Critical Control Points
- **ISO 22000:** Food Safety Management System
- **BRC/SQF:** Global food safety standards
- **Kosher:** Various Orthodox certifications available
- **Halal:** Islamic certification available

PACKAGING & TRANSPORTATION

Packaging Options

- **Small Scale:** 20kg cartons with polyethylene liner
- **Medium Scale:** 200kg steel drums, food-grade coating
- **Bulk Transport:** Flexitanks, ISO containers
- **Retail Packaging:** Consumer-sized jars and containers

Transportation Requirements

- **Temperature Control:** Maintain stable temperature during transport
- **Insulation:** Protect from temperature fluctuations
- **Documentation:** Full traceability documentation
- **Clean Transport:** Dedicated food-grade transportation

APPLICATIONS GUIDANCE

Chocolate Applications

- **Compound Chocolate:** 15-30% replacement for cocoa butter
- **White Chocolate:** Up to 5% addition for texture improvement
- **Coating Applications:** 20-40% of total fat content
- **Tempering:** Follow standard chocolate tempering procedures

Confectionery Applications

- **Hard Candies:** 1-3% for improved mouthfeel
- **Caramels:** 5-10% for texture enhancement
- **Cream Centers:** 10-20% for stability and richness
- **Baked Goods:** 15-25% replacement for other fats

Processing Tips

- **Melting Temperature:** Keep below 60°C to preserve quality
- **Mixing:** Ensure complete incorporation at liquid temperature
- **Cooling:** Allow controlled cooling for proper crystallization
- **Storage:** Store finished products in cool, dry conditions

TECHNICAL SUPPORT

Application Development

- **Recipe Formulation:** Technical assistance for product development
- **Process Optimization:** Guidance on processing parameters
- **Quality Troubleshooting:** Problem-solving support
- **Shelf-life Extension:** Stability improvement strategies

Documentation & Compliance

- **Certificate of Analysis:** Detailed testing results with each batch
- **Nutritional Labeling:** Support for regulatory compliance
- **Allergen Information:** Comprehensive allergen assessment

- **Regulatory Updates:** Current regulation compliance information
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This specification meets international food safety standards including FDA, EU, and Codex Alimentarius requirements. Regular updates ensure compliance with evolving regulatory frameworks.